

8.8 Modeling With Quadratics in the Real World

Lesson Introduction

 Benchmarks	MA.912.AR.3.6 Given an expression or equation representing a quadratic function, determine the vertex and zeros and interpret them in terms of a real-world context. MA.912.AR.3.8 Solve and graph mathematical and real-world problems that are modeled with quadratic functions. Interpret key features and determine constraints in terms of the context. MA.912.AR.9.6 Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or nonviable options.		 Learning Targets	- I can interpret key features of quadratic relationships in a real-world context. - I can interpret solutions as viable or nonviable options given a real-world context. - I can represent constraints of a real-world context using set-builder notation.	
 Benchmark Coherence	Building On N/A		Working Toward MA.912.AR.9.7 Given a real-world context, represent constraints as systems of linear and non-linear equations or inequalities. Interpret solutions to problems as viable or non-viable options.		
 Essential Questions	- In what ways can key features be interpreted within a given context? - How can I determine if a key feature is viable?				
 Lesson Narrative	In this lesson, students extend their knowledge of quadratic equations and key features in a mathematical context to modeling using quadratics in real-world situations. Students spend time identifying key features and interpreting them within the context of the real-world situation. Students will learn how to apply what they've learned about quadratics and key features in the prior lessons of this unit to represent real-world scenarios.				
 Component Summary	8.8.1 Warm-Up (~10 min.)	Complete an error analysis involving rewriting a quadratic equation into vertex form (<i>SE p. 48</i>)	8.8.2 Guided Instruction (30-35 min.)	Modeling with quadratic relationships in real-world contexts (<i>SE p. 48</i>)	
	8.8.3 Wrap-Up (~5 min.)	Ticket Out the Door (<i>SE p. 51</i>)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Vertex Form - Perfect Square Trinomial - Constraints		N/A		N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may not recall how to identify the key features. - Students may not recognize a value that is not viable for a given context. - Students may misinterpret the problem and the key features given values represent.	8.8.2 Guided Instruction drives the learning in this lesson. Consider how your students learn best when planning. All 18 questions act as a journey of learning that provides opportunities for both teacher modeling and partner work.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
 Differentiate Instruction	Three Reads This lesson deals primarily with real-world word problems. Consider leveraging the Three Reads protocol for each word problem. Pose the following questions before each reading: (1 st reading) What is this situation about? (2 nd reading) What can be counted or measured? (3 rd reading) What are some ways we might solve this?	MA.K12.MTR.4.1: Discussion and Reflection 8.8.1 Warm-Up has students complete an error analysis for a student working on rewriting a quadratic equation into vertex form by completing the square. Students must recognize the error(s) made in the process and identify how to correctly complete the task.
	Word-Problem Strategies Consider providing learning strategies for parsing and decoding word problems. You can model the strategy using the first example from 8.8.2 Guided Instruction and prompt students to use this strategy for the 8.8.3 Wrap-Up.	
	- Four-Column Method: Use columns to name components of parsing a word problem (Column 1: What is the question asking?). - Three Reads: Prompt students to identify specific components during each read (see Literacy and Language Routines). - Decoding: Use your own classroom strategies or partner with an English teacher to use decoding strategies students are already familiar with from prior lessons or classes.	
Lesson Notes:		

8.8.1 Warm-Up: Error Analysis

Component Overview: Students will complete an error analysis involving rewriting a quadratic equation into vertex form. This activity is designed to be done independently.



8.8.1 Warm-Up: Error Analysis

1. Jarvis' work for rewriting the expression $4x^2 + 32x - 67$ in vertex form is shown. Camila told Jarvis he made a mistake. Find Jarvis' mistake and correct it.

$$(4x^2 + 32x) - 67$$

$$4(x^2 + 8x) - 67$$

$$4(x^2 + 8x + 16) - 67 - 16$$

$$4(x + 4)^2 - 83$$

Javis did not multiply 16 by 4. It should be:

$$4(x^2 + 8x + 16) - 67 - 64$$

$$4(x + 4)^2 - 131$$



For 8.8.1 Warm-Up, try not to provide substantial instruction for this problem. The idea of this activity is for students to recall what they learned in the previous lessons. This can, however, give you an opportunity to see what they remember and provide clarification in the process as you see any errors.

8.8.2 Guided Instruction: Modeling With a Quadratic Relationship

Component Overview: Students explore a quadratic relationship in real-world context. Students work to identify key features, explain these features in context, and represent both key features and the quadratic using a coordinate plane.



8.8.2 Guided Instruction: Modeling with a Quadratic Relationship

Greenlings are a group of fish that share common features. The function $G(x) = -10x^2 + 100x + 44$ models the number of greenlings caught, in thousands, in the United States, since 2006.



1. What does the variable x represent? Include the units.
 x represents the number of years since 2006.
2. What does the variable G represent? Include the units.
 G represents the number of fish caught, in thousands, since 2006.
3. Find $G(2)$.
 $G(2) = 204$
4. Interpret $G(2)$. Include units.
 $G(2)$ represents 204,000 fish caught in two years after 2006.
5. What is the y -intercept?
 $(0, 44)$
6. Interpret the y -intercept. Include units.
In 2006 (0 years), 44 thousand fish were caught.
7. Rewrite the function in vertex form. Show your work.
 *$G(x) = (-10x^2 + 100x) + 44$
 $G(x) = -10(x^2 - 10x) + 44$
 $G(x) = -10(x^2 - 10x + 25) + 44 + 250$
 $G(x) = -10(x - 5)^2 + 294$*



Leverage word-problem strategies like the Four-Column method, Three Reads, or Think Aloud based on the needs of your students. Consistently provide opportunity for students to make sense of information with a partner during instruction (i.e., during questions 11-12).



Utilize inquiry during this activity.

- Do you think that the function could model the number of Greenlings in the year 2000? In the year 2090?
- What x -values make sense?
- What y -values make sense?
- What year is two years after 2006?

8.8.2 Guided Instruction: Modeling With a Quadratic Relationship (continued)

8. Interpret the vertex in terms of the context. Include units.

Answers may vary. The vertex, (5,294), represents 5 years after 2006, 294,000 fish were caught.

9. Complete the table of values for $G(x)$.

x	-3	0	2	3	5	8	10	11
G	-346	44	204	254	294	204	44	-66

10. Circle the values in the table that do not make sense in terms of the context.

Answers circled on table above.

11. Explain why these values do not make sense.

Answers will vary. You can't catch a negative amount of fish, and you can't have a negative amount of years.

12. Work with your partner to complete as many of the key features as possible using the given function $G(x)$, the function rewritten in vertex form, and the table of values. Do not use approximate or rounded values.

Key Features					
x -intercept		vertex	(5,294)	domain	R
x -intercept		increasing	$x < 5$	range	$y \leq 294$
y -intercept	(0,44)	decreasing	$x > 5$	opens	Down
positive		left end behavior	$x \rightarrow -\infty, y \rightarrow -\infty$		
negative		right end behavior	$x \rightarrow \infty, y \rightarrow -\infty$		



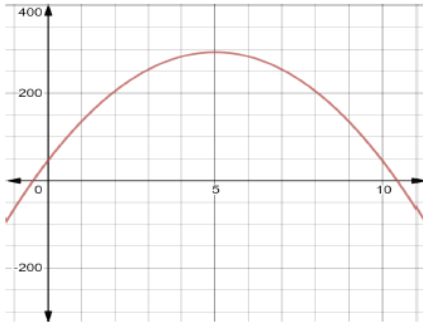
Students may assume that any negative value could not be possible in the context of time. Prompt students to explain the difference between “negative years” and “years prior” as an important distinction when interpreting this context.



- Why were you unable to use this equation to find the intercepts and where the quadratic was positive and negative?*
- What could you do, without graphing, in order to identify those key features?*

8.8.2 Guided Instruction: Modeling With a Quadratic Relationship (continued)

13. Graph the function.



14. Use the graph to estimate the x -intercepts.

The x -intercepts are about $(-0.5, 0)$ and $(10.5, 0)$ or similar estimations.

15. Use the x -intercepts to determine when the number of greenlings caught reaches 0.

Answers will vary based on estimations from question 14. The number of greenlings caught will reach 0 in about 10.5 years.

16. With your partner discuss whether it's reasonable for the number of greenlings caught to be 0. Summarize your discussion.

Answers will vary. The number of greenlings caught may not never hit zero. This would likely only happen if greenlings become extinct.

17. Use the graph, table of values, or functions to determine the reasonable constraints for the context. Write the constraints in set-builder notation.

$\{x | 0 \leq x \leq 10\}$. Students may also choose to use their estimated x -intercept in place of 10.

18. Discuss with your partner why these constraints may be different than the domain of the function

$$G(x) = -10x^2 + 100x + 44.$$

Answers will vary. The domain contains all the possible x -values for the function, but the constraints are the reasonable values for x in context of the problems.



Provide a visual entry-point by showing question 13 not only by hand but also by using an online graphing utility. If you have the appropriate technology to do so, display the graph on a large screen and have students locate the key features they identified earlier on the graph of this question.



Consider highlighting student responses whole group to clarify misconceptions to better prepare students for the 8.8.3 Wrap-Up.

8.8.3 Wrap-Up: Ticket Out the Door

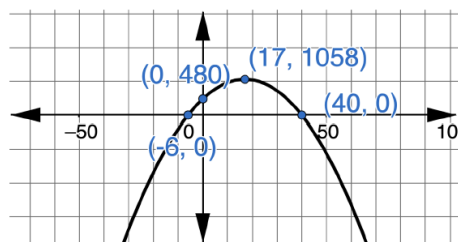
Component Overview: Students will complete a Ticket Out the Door involving interpreting key features of a quadratic equation with a given real-world context. This activity is designed to be done independently in order to provide accurate student-level data.



8.8.3 Wrap-Up: Ticket Out the Door

1. A statistician gathers data over several years, starting in 1995, on the average number of trips a household makes to run errands per year. The statistician uses the data to find an equation that models the average annual number of trips for errands per household for the years since 1995. The statistician determined the function $E(x) = -2x^2 + 68x + 480$ models the situation.

The graph of the function is shown.



- a. Interpret $E(17)$. Include units.
 $E(17)$ represents the number of trips a household makes to run errands 17 years after 1995. 17 years after 1995 is 2012. Therefore, in 2012 a household makes 1058 trips to run errands.
- b. The x -intercepts are labeled on the graph. Determine how reasonable the x -intercepts are by explaining what they each mean in terms of when the average annual number of trips for errands per household will reach 0.
Answers will vary. The two x -intercepts are $(-6, 0)$ and $(40, 0)$. $(40, 0)$ is reasonable because in 40 years, households could be making zero trips to the stores. This could be possible if everyone orders things online. $(-6, 0)$ may be possible if the data supports the model for 1989.



Consider including certain prompts for each portion depending on your students.

- **Question 1a:** Students should identify and interpret both the value and meaning of years and number of trips.
- **Question 1b:** Students should identify the distinction between “There cannot be negative years” and “This is possible, as this would represent six years prior to 1995. However, it would not be reasonable to use for this context because we only need to know since 1995.”